Here $\chi_E = I_E$ or 1_E (indicator function)

2 Lebesgue integration

1. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. We will always assume that μ is complete, otherwise we first take its completion. The example to have in mind is the Lebesgue measure on $\mathbf{R}^n$, $(\mathbf{R}^n, \mathcal{L}_n, |\cdot|)$. We will build the integration theory for $\mathcal{A}$-measurable functions. We will consider measurable functions
$$f\colon \Omega \longrightarrow \bar{\mathbf{R}},$$
where $\bar{\mathbf{R}} = \mathbf{R}^1 \cup \{-\infty\} \cup \{+\infty\}$ (and also functions $F\colon \Omega \to \bar{\mathbf{C}} \cup \{\infty\}$, where $\bar{\mathbf{C}} = \mathbf{C} \cup \{\infty\}$). First we define integrals of real valued nonnegative functions, and then reduce the general case to this. We will follow Rudin very closely.

2. Let $f\colon \Omega \to \mathbf{R}$ be a *simple function*:
$$f = \sum_{j=1}^{N} c_j \chi_{E_j},$$
$c_{1,\dots,N} \in \mathbf{R}$, $E_{1,\dots,N} \in \mathcal{A}$, $N < +\infty$.
It is useful to know that any simple f can be written as a linear combination of χ_{E_j}, $E_j \in \mathcal{A}$, with *distinct* $c_1, \dots, c_N$ and *disjoint* $E_1, \dots, E_N$. In fact, a simple f takes only a finite number of distinct values $\alpha_1 < \alpha_2 < \dots < \alpha_M$. Hence the desired representation is
$$f = \sum_{m=1}^{M} \alpha_m \chi_{f^{-1}(\{\alpha_m\})}.$$
Adding a term with $c_j = 0$ if necessary, we can always assume that
$$\bigcup_{j=1}^{N} E_j = \Omega.$$

3. Let $f \geq 0$ be a simple function written as
$$f = \sum_{n=1}^{N} c_n \chi_{E_n}, \quad E_j \cap E_k = \emptyset \text{ if } j \neq k.$$
Define the Lebesgue integral of f with respect to μ over Ω by
$$\int_\Omega f(x)\, d\mu(x) = \int_\Omega f\, d\mu \stackrel{\text{def}}{=} \sum_{j=1}^{N} c_j \mu(E_j)$$
agreeing that
$$0\infty = 0.$$

We do not assume here that $c_{1,\dots,N}$ are distinct. (However, since E_j s are disjoint, we have $c_{1,\dots,N} \geq 0$.) The definition is correct. In fact, suppose we also have

$$f = \sum_{m=1}^{M} d_m \chi_{F_m}, \quad F_j \cap F_k = \emptyset \text{ if } j \neq k.$$

Notice that

$$\chi_{A \cup B} = \chi_A + \chi_B \quad \text{if} \quad A \cap B = \emptyset.$$

But then

$$\begin{aligned} f &= \sum_{n=1}^{N} \sum_{m=1}^{M} e_{nm} \chi_{E_n \cap F_m}, \\ & e_{nm} = c_n = d_m. \end{aligned}$$

Now, keeping in mind that all sets E_n (and all F_m) are disjoint we derive

$$\begin{aligned} \sum_{n=1}^{N} c_n \mu(E_n) &= \sum_{n=1}^{N} c_n \sum_{m=1}^{M} \mu(E_n \cap F_m) \\ &= \sum_{n=1}^{N} \sum_{m=1}^{M} e_{nm} \mu(E_n \cap F_m), \end{aligned}$$

and by the same argument

$$\sum_{m=1}^{M} d_m \mu(F_m) = \sum_{n=1}^{N} \sum_{m=1}^{M} e_{nm} \mu(E_n \cap F_m).$$

So $\int f \, d\mu$ is well defined.

ex: $f: \mathbb{R} \to \mathbb{R}$

$$\begin{aligned} f(x) &= 2 \cdot 1_{[0,2)} + 3 \cdot 1_{[1,4)} \\ &= 2 \cdot 1_{[0,1)} + 5 \cdot 1_{[1,2)} + 3 \, 1_{[2,4)} \\ &= 2 \cdot 1_{[0,1)} + 5 \cdot 1_{[1,2)} + 3 \cdot 1_{[2,3)} + 3 \cdot 1_{[3,4)} \\ &\quad + 0 \cdot 1_{[[0,4)^c)} \end{aligned}$$